EXTENDING A FINITE TYPE QUASI-COHERENT SUBSHEAF FROM A QUASI-COMPACT OPEN

THE STACKS PROJECT AUTHORS

Editorial note. All delivered mathematical text is the author's own, copied from the pinned Stacks Project chapter file properties.tex; no statement or proof step was rewritten, reordered or removed.

Editorial note. The counted proof is the author's complete argument: the reduction to one affine piece outside U and then to an affine base by glueing, the production of a quasi-coherent subsheaf restricting to the given one, and the cut-down to finite type by covering U with finitely many localizations where the module is finitely generated.

Editorial note. The label lemma-extend names the counted statement, and the proof environment immediately following it in the source is delivered verbatim as its proof.

Editorial note. The counted statement is carried by the author's own lemma environment in the chapter source.

Editorial note. Pre-open gate: python operations/agent-authoring/tools/precard_gate.py –source properties.tex –label lemma-extend –start 3063 –end 3112 –local-id 28035 returned RESULT=OK (L28035 free, occupancy FREE, public interval clear). Candidate taken from sources/stacks-properties/candidates-halfA.json.

Editorial note. The statement and proof are copied from the pinned chapter file; the cited inputs are kept as external pointers with their canonical Stacks tags.

Allowed standard prerequisites (editorial).

- Trivial extension of a quasi-coherent subsheaf from an open of an affine scheme. Same chapter, the lemma producing a quasi-coherent subsheaf of the given module restricting to the given subsheaf over the open. Supplies the subsheaf H of F restricting to G over U in the affine case.
- Finite generation of a localization at a point of a quasi-coherent subsheaf. Same chapter, the lemma that a quasi-coherent module restricting to a finite type module over a quasi-compact open is generated by finitely many elements after localizing at suitable elements. Supplies, for each point of U, an element f with N_f finitely generated and D(f) contained in U.

Source reference. Original reference 01PE: sources/stacks-properties/pinned-original-source.tex, source lines 3011–3011. Prerequisite pointer invoked at proof line(s) 3098; it names a standard fact of the same chapter that the counted argument uses, and every step of the counted proof built on it is retained verbatim in the delivered body.

Source reference. Original reference 01PB: sources/stacks-properties/pinned-original-source.tex, source lines 2310–2310. Prerequisite pointer invoked at proof line(s) 3104; it names a standard fact of the same chapter that the counted argument

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uses, and every step of the counted proof built on it is retained verbatim in the delivered body.

Lemma 0.1. *Let X be a quasi-compact and quasi-separated scheme. Let $U \subset X$ be a quasi-compact open. Let $\mathcal{F}$ be a quasi-coherent $\mathcal{O}_X$-module. Let $\mathcal{G} \subset \mathcal{F}|_U$ be a quasi-coherent $\mathcal{O}_U$-submodule which is of finite type. Then there exists a quasi-coherent submodule $\mathcal{G}' \subset \mathcal{F}$ which is of finite type such that $\mathcal{G}'|_U = \mathcal{G}$.*

Proof. Let n be the minimal number of affine opens $U_i \subset X$, $i = 1, \ldots, n$ such that $X = U \cup \bigcup U_i$. (Here we use that X is quasi-compact.) Suppose we can prove the lemma for the case $n = 1$. Then we can successively extend $\mathcal{G}$ to a $\mathcal{G}_1$ over $U \cup U_1$ to a $\mathcal{G}_2$ over $U \cup U_1 \cup U_2$ to a $\mathcal{G}_3$ over $U \cup U_1 \cup U_2 \cup U_3$, and so on. Thus we reduce to the case $n = 1$.

Thus we may assume that $X = U \cup V$ with V affine. Since X is quasi-separated and U, V are quasi-compact open, we see that $U \cap V$ is a quasi-compact open. It suffices to prove the lemma for the system $(V, U \cap V, \mathcal{F}|_V, \mathcal{G}|_{U \cap V})$ since we can glue the resulting sheaf $\mathcal{G}'$ over V to the given sheaf $\mathcal{G}$ over U along the common value over $U \cap V$. Thus we reduce to the case where X is affine.

Assume $X = \mathrm{Spec}(R)$. Write $\mathcal{F} = \widetilde{M}$ for some R-module M. By Lemma 01PE above we may find a quasi-coherent subsheaf $\mathcal{H} \subset \mathcal{F}$ which restricts to $\mathcal{G}$ over U. Write $\mathcal{H} = \widetilde{N}$ for some R-module N. For every $u \in U$ there exists an $f \in R$ such that $u \in D(f) \subset U$ and such that N_f is finitely generated, see Lemma 01PB. Since U is quasi-compact we can cover it by finitely many $D(f_i)$ such that N_{f_i} is generated by finitely many elements, say $x_{i,1}/f_i^N, \ldots, x_{i,r_i}/f_i^N$. Let $N' \subset N$ be the submodule generated by the elements $x_{i,j}$. Then the subsheaf $\mathcal{G}' = \widetilde{N'} \subset \mathcal{H} \subset \mathcal{F}$ works. $\square$